



CARGO CONVOY CRM

User Manual & System Documentation

Cargo Convoy Inc

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Version 2.1 | July 2026

Built with Laravel 12 | PHP 8.4

Routes: 305 | Controllers: 15 | Models: 34 | Views: 123

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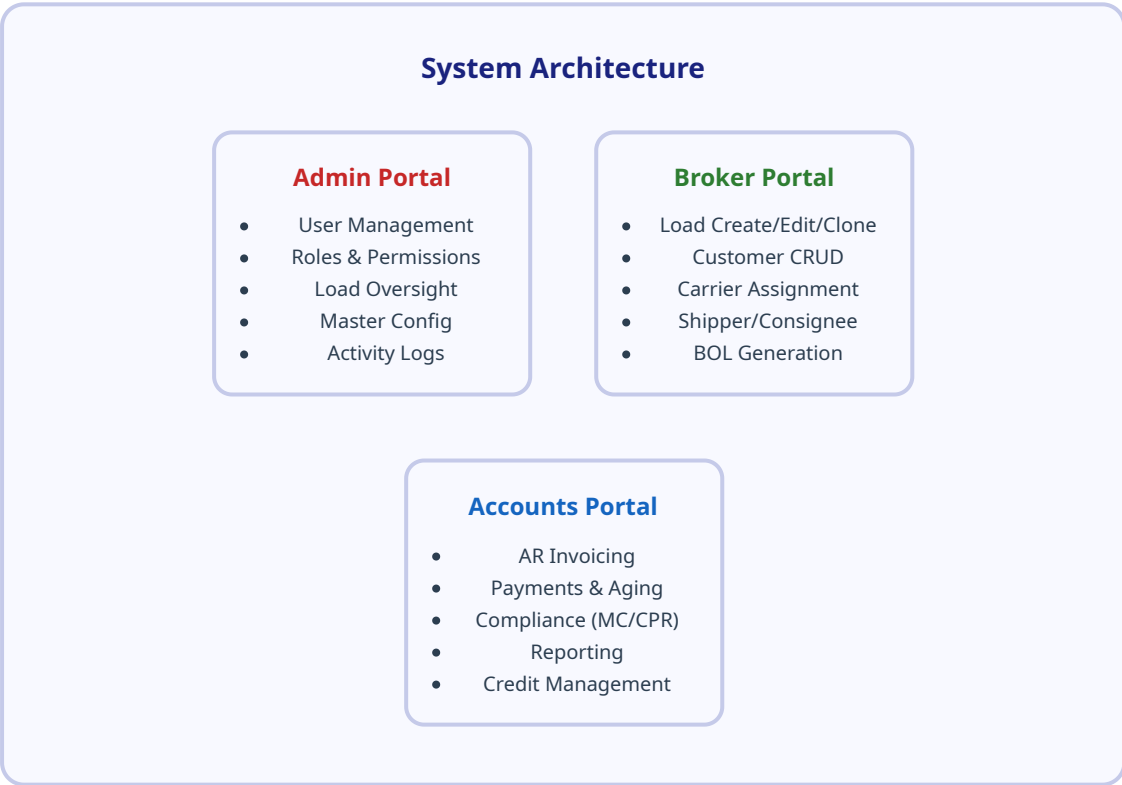
1. System Overview & Architecture

1.1 What is Cargo Convoy CRM?

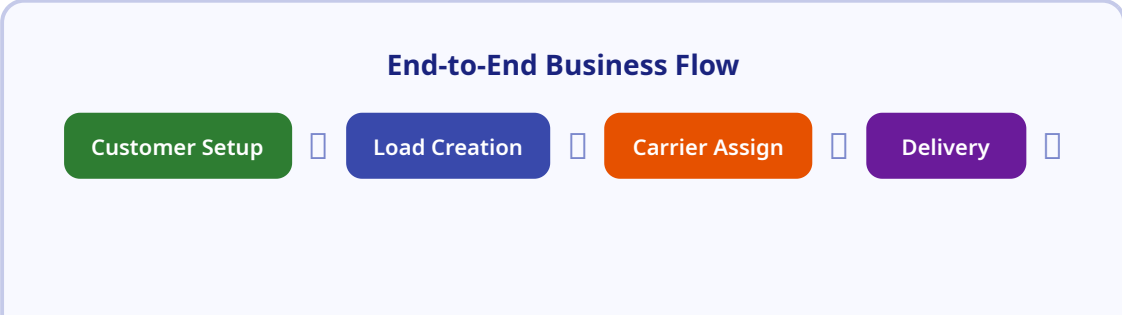
Cargo Convoy CRM is a freight brokerage management system for **Cargo Convoy Inc.** It manages the full lifecycle of freight loads — from customer setup, load creation, carrier dispatch, through to invoicing and payment collection.

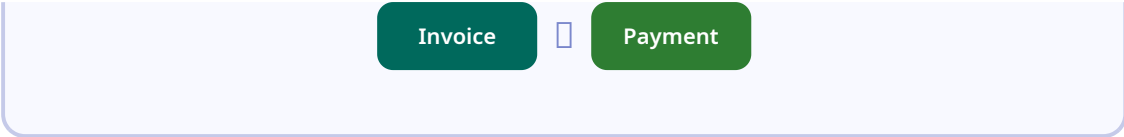
Tech Stack: Laravel 12 (PHP 8.4) | MySQL | Blade Templates | Bootstrap 5 | jQuery | Select2 | Dompdf | FPDF | PhpSpreadsheet

1.2 Three-Portal Architecture



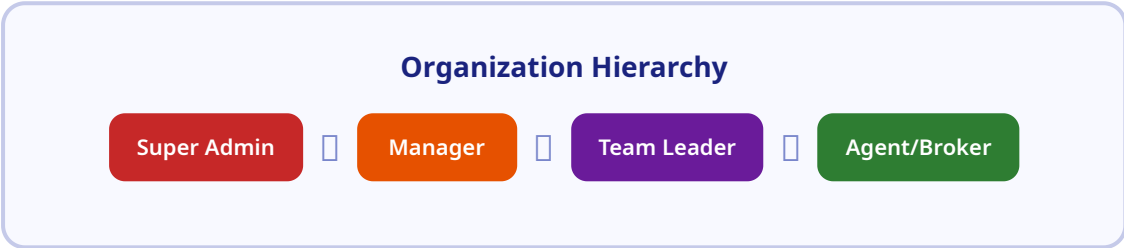
1.3 High-Level Process Flow





2. User Roles & Access Control

2.1 Role Hierarchy



2.2 Access Matrix

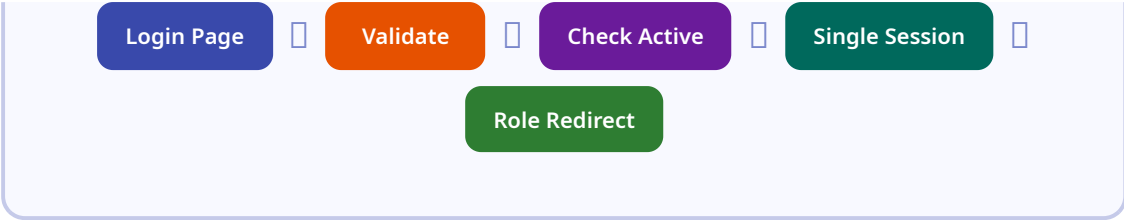
Role Group	IDs	Portal Access	Landing Page
Admin	1, 2, 3, 22	Admin + Broker + Account	/admin/admin_home
Accounting	7, 8, 9, 24	Account + Broker	/account/accounting
Vendor/AP	4, 5, 6	Account + Broker	/account/vendor-system
Compliance	10, 11, 12	Account + Broker	/account/compliance
Manager	19	Broker	/broker/load
Team Leader	20	Broker	/broker/load
Agent	21	Broker	/broker/load

2.3 Data Visibility

Admins: See ALL data | **Managers:** See their TLs + agents | **Team Leaders:** See their agents | **Agents:** Own data only

3. Authentication & Login



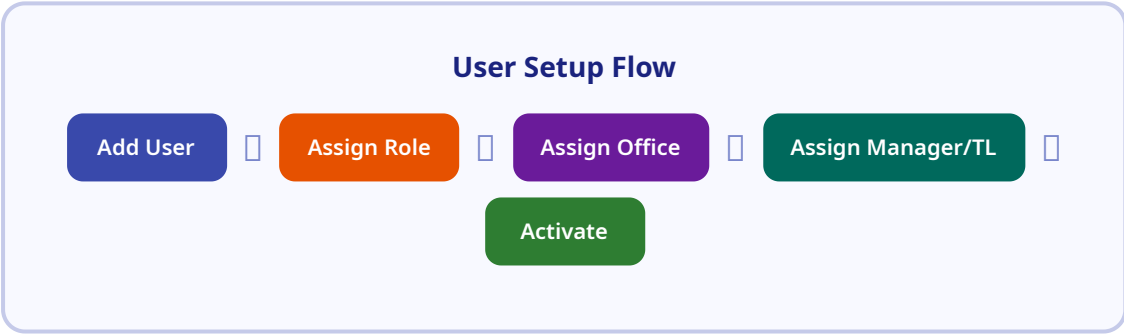


Security Features

- 1 **Single Session:** Only one active login per user. Second device is blocked.
- 2 **Active Check:** Only users with status="active" can login.
- 3 **Activity Log:** Every login/logout is recorded with IP and timestamp.

4. Admin Panel

4.1 User Management



4.2 Organization Structure

Entity	Operations	Purpose
Departments	CRUD	Broker, Account, Admin, IT
Offices	CRUD	Physical locations
Managers	CRUD	Link users as managers
Team Leaders	CRUD	Link users under managers

4.3 Master Data

Data	Examples
Equipment Types	Dry Van, Flatbed, Reefer, Step Deck
Shipment Types	FTL, LTL, TONU, Partial

Countries & States

US, Canada + all states/provinces

4.4 Activity Logs

Every action (create, update, delete) is logged with user info, IP address, old/new data in JSON format. Accessible at /admin/activity_logs.

5. Broker Module

5.1 Broker Dashboard

The main workspace for freight brokers. After login, brokers see their loads with status tabs: All, Open, Delivered, Complete, Invoiced, Paid.

5.2 Module Features

Load Management

- Create / Edit / Clone
- Status tracking
- Credit validation

Customer

- Add / Edit
- Credit monitoring
- Remittance upload

Carrier

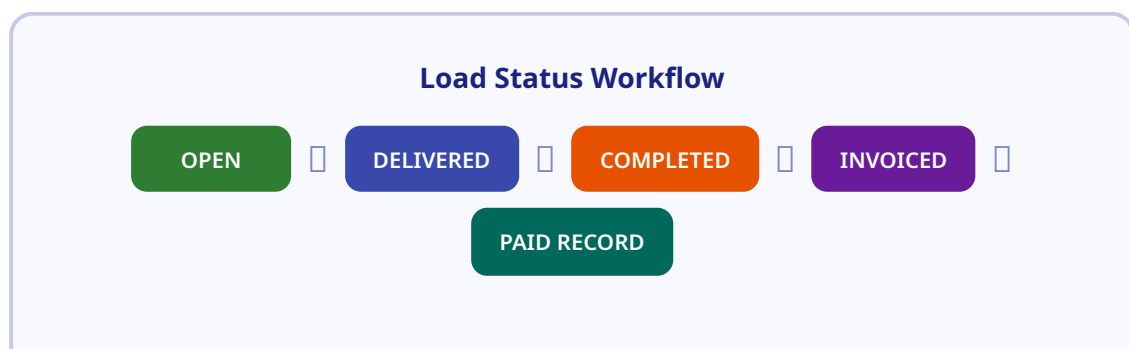
- My Carriers / All
- MC/DOT search
- Auto-suggest

Shipper/Consignee

- CRUD
- Cross-reference
- Auto-fill in loads

6. Load Management (Core Module)

6.1 Load Lifecycle



Special: CANCELLED (any stage) | TONU (Truck Order Not Used)

6.2 Load Creation Steps

- 1 Select Customer (Bill To):** Choose an approved customer. Credit limit validated.
- 2 Set Load Type:** OTR or DRAYAGE.
- 3 Add Shipper(s):** Multiple stops supported (up to 15). Location, appointment, commodity.
- 4 Add Consignee(s):** Delivery must be after pickup date.
- 5 Assign Carrier:** Search by Name/MC#/DOT#. Auto-fills details.
- 6 Set Rates:** Customer Rate + FSC + Charges = Final Rate. Carrier fee must not exceed customer rate.
- 7 Credit Check:** Server-side validation blocks if credit exceeded.

6.3 Rate Calculation

Rate Formula

Customer: Base Rate + (FSC% x Rate) + Other Charges = **Final Customer Rate**

Carrier: Carrier Fee + (FSC% x Fee) + Charges = **Final Carrier Fee**

Profit: Customer Final Rate - Carrier Final Fee = **Margin**

6.4 Load Clone

Clone copies all load data but resets rates to 0 and status to "Open". Saves time for repeat shipments.

7. Credit Limit System

7.1 How Credit Works

Credit Flow Diagram

Credit Assigned

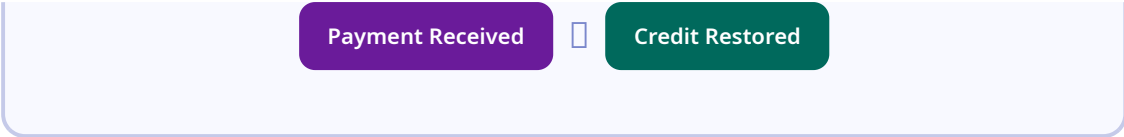


Load Created



Credit Deducted





7.2 Credit Fields

Field	Purpose
adv_customer_credit_limit	Total assigned credit limit
remaining_credit	Available balance (limit - used)
remaining_credit_amount	Synced copy of remaining_credit
invoice_credit_limit	Separate invoice-specific limit
credit_limit_log	JSON audit trail of limit changes

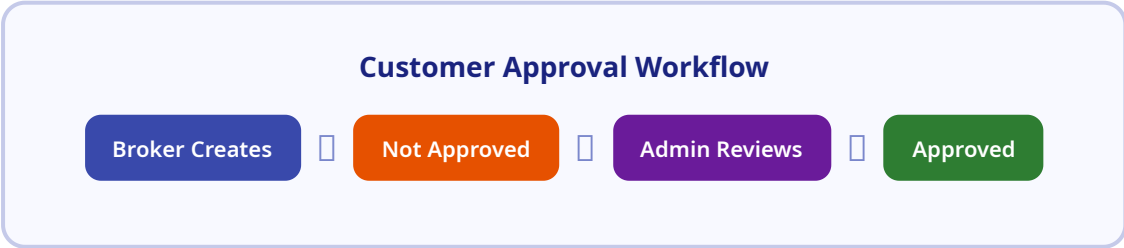
7.3 Credit Validation (Server-Side)

```
// CreditService.php - Atomic check + deduction with row locking
DB::transaction(function() {
    $customer = Customer::lockForUpdate()->find($id);
    $available = calculateAvailableCredit($customer);
    if ($loadAmount > $available) {
        return error("Insufficient credit. Available: $available");
    }
    $customer->remaining_credit -= $loadAmount;
    $customer->save();
});
```

Security: Credit is validated server-side with database row locking. Users cannot bypass by disabling JavaScript or submitting forms directly.

8. Customer Management

8.1 Customer Lifecycle



Important: Only approved customers can be selected for load creation.

8.2 Customer Fields

Section	Fields
Basic	Name, MC/FF#, Corporation, Status
Address	Street, Country, State, City, ZIP
Billing	Separate billing address
Contacts	Primary, Secondary, Billing email
Credit	Credit Limit, Payment Terms, Factoring

8.3 Remittance

Upload remittance files (PDF/images) with timestamps and notes. Filter by date range.

9. Carrier Management

9.1 Carrier Data

Section	Fields
Identity	Name, MC#, DOT#
Contact	Name, Email, Phone, Fax
Business	Payment Terms, Factoring, Load Type
Status	Active / Blacklisted / Blocked

9.2 Carrier Auto-Suggest

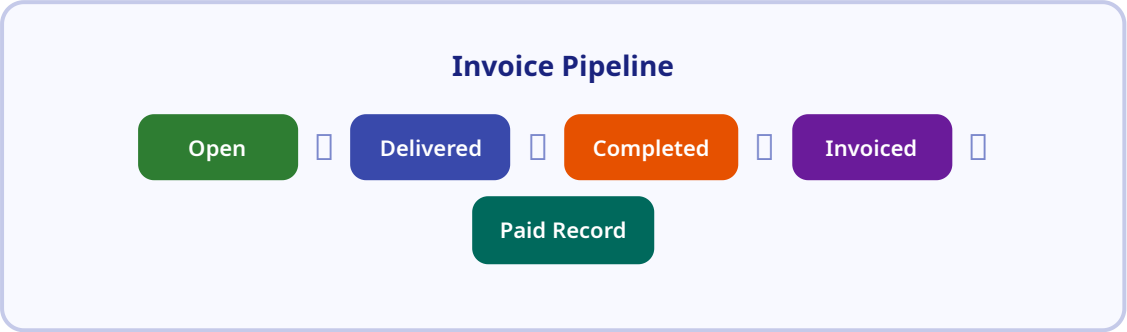
When typing in load form, system provides live suggestions via AJAX. Selecting auto-fills all carrier fields. Carriers must exist in DB and be MC-approved to be assigned.

9.3 Carrier Verification

Accounts team verifies carriers with factoring company checks, bank document uploads, and insurance validation before carriers can be used on loads.

10. Accounting & Invoicing

10.1 AR Accounting Workflow



10.2 Invoice Process

- 1 **Mark Invoiced:** Set invoice_status to 'Paid' (invoiced to customer)
- 2 **Email Invoice:** Send with merged PDF attachments
- 3 **Record Payment:** Enter receiving_amount
- 4 **Paid Record:** Final status when fully paid

10.3 Carrier Payments

Feature	Description
Quick Pay	Mark for expedited payment
Payment Method	ACH, Check, Zelle
Mark as Paid	Record carrier payment
Due Date	Set carrier payment deadline

10.4 AR Aging & Close

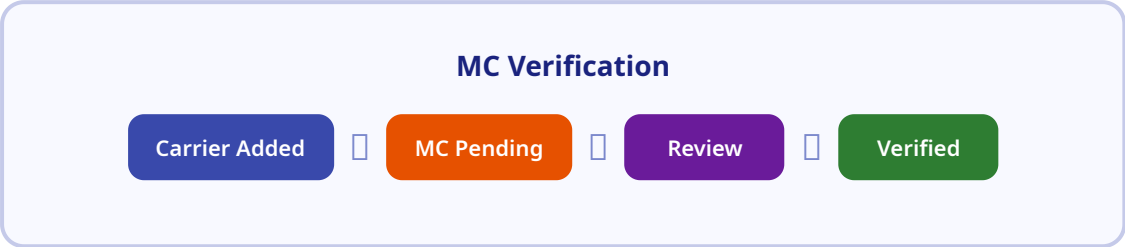
The aging report shows days since invoice. Account team can close aging records when payments are reconciled. Excess/remaining amounts tracked per load.

10.5 All Load Status AR

A comprehensive view showing all loads across all statuses for the AR team, with filtering by office, agent, and date range.

11. Compliance

11.1 MC Check



11.2 CPR Check

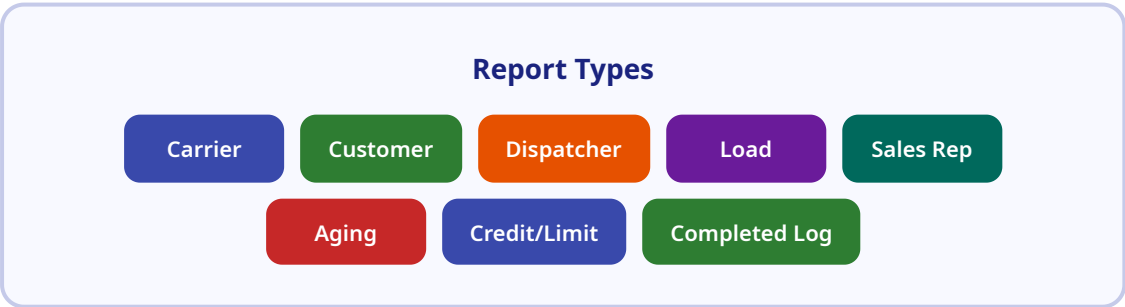
Status	Meaning
Not Approved	Default - not yet reviewed
Verified	Carrier approved for load
Not Verified	Carrier rejected

11.3 Carrier Block

Compliance team can block carriers. Blocked carriers cannot be assigned to new loads.

12. Reporting & Analytics

12.1 Available Reports



12.2 Report Details

Report	Key Metrics	Export
Carrier	Revenue per carrier, load count	Excel

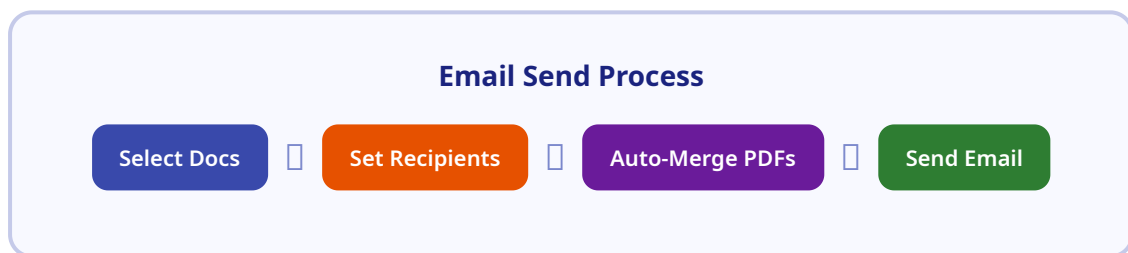
Customer	Revenue, margin, credit usage	Excel
Dispatcher	Revenue per agent, margins	Excel
Aging	Days since invoice, outstanding amounts	Excel
Credit/Limit	Assigned vs remaining credit	Excel

12.3 Sales Dashboard

Interactive chart showing daily revenue trends. Filter by date range, office, and agent. Displays Shipper Rates (revenue), Carrier Fees (cost), and Margin (profit).

13. Email & PDF Merge

13.1 Invoice Email Flow



13.2 PDF Merge

Multiple documents (PDFs + images) are merged into a single PDF attachment using FPDF + Dompdf. Images are auto-converted to PDF before merging.

13.3 Email Template

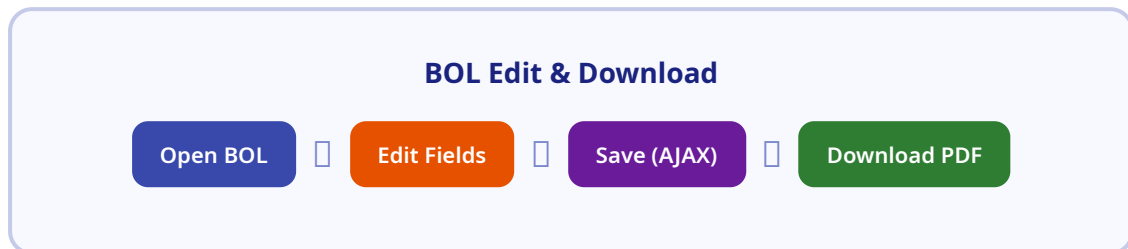
Subject: Invoice For Load #{load} (#{invoice}) REF #{ref}
 From: ar@cargoconvoy.co
 Body: Professional template with company details
 Attachments: Merged PDF of all selected documents

13.4 Features

- Multiple TO recipients (comma-separated)
- CC support
- Custom subject line
- Auto-merge multiple files into one PDF
- Secure: Only files from allowed directories can be attached

14. Bill of Lading (BOL)

14.1 BOL Workflow



14.2 BOL Sections

Section	Content
Header	Company logo, name, address
Load Info	Load#, BOL#, Ship/Delivery dates
Shipper	Name(s) and address(es)
Consignee	Name(s) and address(es)
Freight Table	Pieces, Description, Weight, Class
Signatures	Shipper, Carrier, Consignee with dates

15. Document Management

15.1 Load Document Types

Document	Purpose
Carrier Rate Confirmation	Signed rate agreement with carrier
Proof of Delivery (POD)	Delivery confirmation
Shipper Rate Approval	Customer rate approval
Carrier Invoice	Carrier's invoice for payment
Delivery Order (DO)	Original delivery order
Optional Documents	Any additional supporting docs

15.2 File Storage

```
public/uploads/  
Load-files/{load_number}-{customer_id}/ (Public docs)  
delivery-order/{load_id}/ (Private docs)  
carrier/ (Carrier docs)  
customers/{customer_name}/ (Customer files)  
remittances/ (Remittance files)  
mergedfiles/ (Merged PDFs)
```

16. Database Schema

16.1 Core Tables

Table	Purpose	Key Relations
users	All system users	role, department, office
loads	Freight loads	user, customer, carrier
customers	Customer entities	user; has_many: loads
externals	Carriers	user; has_many: loads
shippers	Shipper locations	user
consignees	Consignee locations	user
roles	User roles	permissions
permissions	System permissions	department
logs	Activity audit trail	All CRUD ops

16.2 Load Table Key Columns

Group	Fields
Identity	id, load_number, user_id, customer_id, carrier_id
Status	load_status, invoice_status, cpr_check
Rates	load_shipper_rate, shipper_load_final_rate, load_final_carrier_fee
Shipper (JSON)	load_shipperr, load_shipper_location, load_shipper_appointment
Invoice	invoice_number, invoice_date, receiving_amount

17. API Endpoints Summary

17.1 Admin Routes

Method	Endpoint	Purpose
GET	/admin/admin_home	Admin dashboard
POST	/admin/create-user	Create user
POST	/admin/update-status	Activate/deactivate
POST	/admin/load/update/{id}	Update load (admin)
GET	/admin/activity_logs	View all logs

17.2 Broker Routes

Method	Endpoint	Purpose
GET	/broker/load	Load listing
POST	/broker/load-create	Create load
POST	/broker/load/update/{id}	Update load
GET	/broker/clone/load/{id}	Clone load
GET	/broker/load/{id}/bol/pdf	BOL PDF
POST	/broker/customer/create	Create customer
POST	/broker/create/carrier	Create carrier

17.3 Account Routes

Method	Endpoint	Purpose
GET	/account/accounting	AR dashboard
POST	/account/send-mail	Send invoice email
POST	/account/update-invoice-status/{id}	Mark invoiced
GET	/account/reporting	Reports
GET	/account/credit	Credit management
GET	/account/compliance	Compliance
POST	/account/update-ar-aging-close	Close aging record

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